

1. Establishing the a priori 1D model(s)

Obtain all available a priori (prior to the 1D or 3D inversion) information regarding the stratification of the area under study (velocities, layer thicknesses, etc.). In general, use refraction seismic models, simplified where necessary to constant velocity layers. If no controlled-source seismology models are available, use phase correlations and cross over distances [e.g., Deichmann, 1987] from well-recorded earthquakes and/or infer the layered structure from geologic information. Define the media by several layers of increasing velocity with depth. Thicknesses of the layers in the upper crust should be about 2 km and in the lower crust about 4 to 5 km. Estimate layer velocities according to a priori information or a general crustal model. In case of incomplete or inconsistent information, or, if the area under consideration confines two or more distinctly different tectonic provinces, establish several 1D models. Choose a reference station with a continuous or nearly-continuous record of events. It must be a reliable station, preferably located toward the center of the network, and should not show extreme site effects. The model(s) and the reference station are called the a priori 1D model(s). If several significantly different a priori 1D models are established the following steps 2 through 5 are repeated for each 1D model separately.

2. Establishing the geometry and the velocity intervals of potential 1D model(s)

Select about 500 of the best events in the data (i.e., those with the most high-quality P-arrivals) that cover the entire area under consideration. Relocate them with routine VELEST using a damping coefficient of 0.01 for the hypocentral parameters and the station delays, and 0.1 for the velocity parameters. Invert for hypocenters every iteration and for station delays and velocity parameters every second iteration. Repeat this procedure several times with new (updated) velocities in the reference 1D model, with perhaps the new station delays, and with new hypocenter locations. Repeat the procedure also for reduced number of layers where possible by combining adjacent layers with similar velocities. Unless clearly indicated by the data, in most cases it is preferable to avoid low velocity layers, as they normally introduce instabilities.

Our experience suggests that shot or blast data should not be included in the 1D-model inversion. Rather, such data should be used to set the near surface velocities, and to test the performance of the resulting minimum 1D model when used for locating hypocenters. This counterintuitive suggestion may be understood by considering that raypaths with both endpoints near the surface sample, on average, a much more heterogeneous part of the Earth than do raypaths from events in the seismogenic crust.

The goal of this trial and error approach is to establish reasonable geometry of the crustal model and corresponding intervals for the velocity parameters and station delays. In addition, this approach provides valuable knowledge about the quality of the data. Proceed to the next step when, a) the earthquake locations, station delays, and velocity values do not vary significantly in subsequent runs; b) the total RMS-value of all events shows a significant reduction with respect to the first routine earthquake locations; and c) the calculated 1D velocity model and the set of station corrections make some geological sense (e.g., stations with negative traveltime residuals should lie in local high velocity areas with respect to the reference station, etc.) and do not

violate a priori information. If all these requirements are satisfied, the result may be called the "updated a priori 1D model with corresponding station residuals".

3. Relocation and final selection of events

Relocate all events using the updated a priori 1D model with station residuals with a routine location procedure (HYPO71 [Lee and Lahr 1975]; HYPOINVERSE [Klein 1978]; HYPOELLIPSE [Lahr 1980]) or with VELEST in the single event mode (fixing the station and velocity parameters). Reselect the best (consider gap, number of observations, distance to next station) 500 or so events that should be well distributed over the volume under investigation. If more than one such subset of about 500 events can be extracted, proceed for each subset separately with step 4 but try to obtain similar results.

4. Calculation of minimum 1D model for one subset

In general terms repeat step 2 with the updated a priori 1D model and station residuals and with a damping of 0.01 for the hypocentral, 0.1 for the station, and 1.0 for the velocity parameters. The goal of this step is to calculate the 1D model (velocity parameters and station residuals), that minimizes the total estimated location errors for a fixed geometry. Test the stability of the result by systematically and randomly shifting hypocenters and by underdamping the velocity parameters. If you are pleased with the performance of the solution fix the updated velocity parameters by overdamping and calculate the station residuals. The resulting velocity model with corresponding station residuals is called "minimum 1D model".

5. Calculation of minimum 1D model for several subsets

If several subsets of 500 events were extracted, test the dependence of your minimum 1D model on specific data. Find the 1D model and station residuals that will best fit the results from all subsets, mix data from different subsets, and repeat step 4. If the results are unsatisfactory, evaluate the best 1D model by the procedure described in step 6.

6. Evaluation of different minimum 1D models for same area

If several significantly different a priori 1D models were established and steps 2 through 5 were successfully completed for each of them, you may base your choice of one minimum 1D model on the result of the following performance test: Select all traveltime data from quarry blasts or shots (i.e., from sources of known location) and relocate these events for the different minimum 1D models without fixing the depth during the location process. If the near surface velocities for several station locations are known, compare the station residuals with the differences between the average layer velocity and the local velocities. Finally, select the minimum 1D model that best resembles the a priori information.